Q1 (20 July 2021 Shift 2)

Two small drops of mercury each of radius R coalesce to form a single large drop. The ratio of total surface energy before and after the change is:

(1) $2^{\frac{1}{3}} : 1$

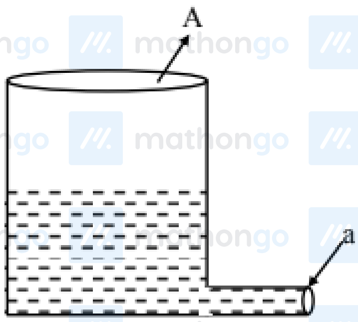
(2) $1 : 2^{\frac{1}{3}}$

(3) $2 : 1$

(4) $1 : 2$

Q2 (27 July 2021 Shift 1)

A light cylindrical vessel is kept on a horizontal surface. Area of base is A. A hole of cross- sectional area 'a' is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to



the impact force of the emerging liquid is $(a<

(1) $\frac{A}{2a}$

(2) None of these

(3) $\frac{2a}{A}$

(4) $\frac{a}{A}$

Q3 (27 July 2021 Shift 2)

A raindrop with radius $R = 0.2$ mm falls from a cloud at a height $h = 2000$ m above the ground. Assume that the drop is spherical throughout its fall and the force of buoyance may be neglected, then the terminal speed attained by the raindrop is :

[Density of water $f_{\mathrm{w}}=1000 \mathrm{~kg} \mathrm{~m}^{-3}$ and Density of air $f_{\mathrm{a}}=1.2 \mathrm{~kg} \mathrm{~m}^{-3}, \mathrm{~g}=10 \mathrm{~m} / \mathrm{s}^{2}$

Coefficient of viscosity of air $=1.8 \times 10^{-5} \mathrm{Nsm}^{-2}$]

(1) $250.6 \mathrm{~ms}^{-1}$

(2) $43.56 \mathrm{~ms}^{-1}$

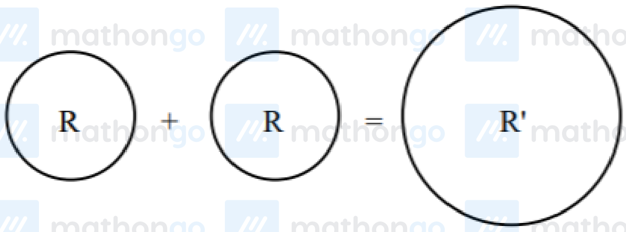
(3) $4.94 \mathrm{~ms}^{-1}$

(4) $14.4 \mathrm{~ms}^{-1}$

Answer Key

Q1 (1) **Q2** (3) **Q3** (3)

Q1



$\frac{4}{3}\pi R^3 + \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R'^3$

$R' = 2^{\frac{1}{3}} R \ldots \text{(i)}$

$\mathrm{A_i} = 2\left[4\pi \mathrm{R}^2\right]$

$\mathrm{A_f} = 4\pi \mathrm{R}'^2$

$\frac{\mathrm{U_i}}{\mathrm{U_f}} = \frac{\mathrm{A_i}}{\mathrm{A_f}} = \frac{2\mathrm{R}^2}{2^{2/3}\mathrm{R}^2} = 2^{1/3}$

Q2

For no sliding

$\mathrm{f} \geq \rho \mathrm{av}^2$

$\mu \mathrm{mg} \geq \rho \mathrm{av}^2$

$\mu \rho \mathrm{Ahg} \geq \rho \mathrm{a2gh}$

$\mu \geq \frac{2a}{A}$

Option (3)

Q3

At terminal speed

$$
\begin{aligned}
a &= 0 \\
F_{net} &= 0 \\
mg &= F_v = 6\pi\eta Rv \\
v &= \frac{mg}{6\pi Rv} \\
v &= \frac{\rho_w \frac{4\pi}{3} R^3\ g}{6\pi\eta R} \\
&= \frac{2\rho_w R^2\ g}{9\eta} \\
&= \frac{400}{81}\ m/s \\
&= 4.94\ m/s
\end{aligned}
$$